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Optimization and parametric studies of two-stage-to-orbit liquid oxygen/paraffin hybrid rocket vehicles

Troy L. Messinger *, Matthew S. Corbiell, Craig T. Johansen *

Mechanical Engineering, University of Calgary, AB T2L 1Y6, Canada

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ABSTRACT

A framework was developed to perform conceptual design studies of hybrid rocket launch vehicles. The framework generates vehicle designs considering the propulsion, structures, aerodynamics, and trajectory/guidance sub-disciplines. The launch vehicles investigated use paraffin wax and liquid oxygen propellants. The payload fraction was assessed for vehicles with either an electric pump-fed or pressure-fed system. A particle swarm optimization found optimal two-stage-to-orbit configurations for both feed systems investigated. Electric pump-fed systems had a payload fraction increase over pressure-fed systems. A parametric study of the vehicle design variables provides verification of finding the local optimum and sensitivity of the payload fraction to change in each design variables.

1. Introduction

Hybrid rockets have the potential to disrupt the launch industry by offering relatively simple mechanical design and lower cost at comparable performance to existing bi-propellant liquid rocket technology [1–3]. Hybrid rockets have the benefit of being inherently safer and mechanically simpler when compared to solid and liquid bi-propellant propulsion systems. Despite several advantages over the competing technologies, all commercial orbital launch vehicle ventures based on it have failed so far. Recently, since the mid-90s, interest in hybrid propulsion has begun to rise. The rising interest is attributed in part to the improvements in the regression rates of the fuel with the use of liquefying fuels [4]. The use of liquefying fuels has made simple, single-port configurations more feasible and reduced the required fuel burning surface area. Recently, the small-satellite market has grown and many startups are competing to offer economical, dedicated small-satellite launch services. Many small-satellite launch vehicle developers are pursuing the use of hybrid rocket technology due to its advantages [5].

This study focuses on the conceptual multi-disciplinary design and optimization (MDO) of hybrid rocket vehicles. The disciplines considered include: propulsion, structures, aerodynamics, and trajectory guidance and navigation. MDO is as a design process that takes into account the mutually interacting phenomena of different disciplines [6]. The use of MDO can find better performing configurations, that can't be found by optimizing a single discipline at each sequential design step. The MDO process can lead to superior optimal results, however it can greatly increase the complexity of the design problem.

Research relating to launch vehicle MDO contains analyses of large payload (> 1000 kg) class, liquid bi-propellant and solid rocket, expendable launchers optimized based on payload fraction [7–11]. Cost and reliability models have been formulated and implemented into the launch vehicle MDO process [12]. Later researchers worked to review, refine, and add fidelity to the engineering models to generate detailed results [9]. Parallel to the engineering model refinement, some researchers focused on the MDO process rather than the engineering models [13,14]. Recent work incorporated re-usability into the launch vehicle conceptual design and optimization process [10,11]. Much of the work on hybrid rocket conceptual design and optimization has focused on the propulsion system specifically [15,16], with less details relating to the entire vehicle and trajectory design. A more comprehensive study on hybrid rocket vehicle and trajectory MDO has been put forward [17]. That study generated optimal three-stage vehicle concepts launched from ground or in-air carrier vehicles but did not investigate potentially high performing propellants: paraffin wax and liquid oxygen [18]. One study found three-stage-to-orbit, pressure-fed, liquid oxygen / paraffin hybrid rockets are viable and had a payload fraction 1% of gross takeoff mass [19].

MDO has been shown to be a powerful tool in producing high-performing launch vehicle concepts. However, many studies have used empirical mass estimating relations (MERs) used for large-

* Corresponding authors.
E-mail addresses: tlmessin@ucalgary.ca (T.L. Messinger), johansen@ucalgary.ca (C.T. Johansen).

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Abbreviations

ASL	Above Sea Level
CEA	Chemical Equilibrium with Application
DOF	Degree of Freedom
IDF	Inter-Disciplinary Feasible
L-BFGS-B	Limited memory Broyden–Fletcher–Goldfarb–Shanno Bounded
MDO	Multi-disciplinary Design Optimization
MER	Mass Estimating Relationship
NPSP	Net Positive Suction Pressure
PSO	Particle Swarm Optimization
SECO-1	Second Engine Cut-Off 1

scale vehicles, which are not effective for small-scale vehicle design. As such, research on small vehicle conceptual design should use physics-based formulations given the lack of historical data. This study aims to address this gap by using a physics-based structural design methodology and only resort to MERs when necessary. This study presents optimal particle swarm hyper-parameters, determined through a grid search. Particle swarm hyper-parameter optimization is a process often omitted in past works. Previous studies often presented optimal design values without a sensitivity or parametric analyses. It is important to conduct a parametric study to show the relative importance of each design parameter to the overall system performance and to verify that the converged solution is at least locally optimal. The objectives of this study are to determine the optimal design values for pressure-fed and electric pump-fed hybrid rocket orbital launch vehicles through a comprehensive conceptual design process, and to assess the sensitivity of these design parameters on the global performance through a parametric study. This study only considers liquid oxygen due to its high performance as an oxidizer and paraffin wax fuel due to its high regression rate.

1.1. Propellant feed systems

Two main oxidizer feed systems considered in the present study are electric pump-fed and gas pressure-fed (i.e. backfill). The backfill systems use a pressurizing system comprised of a separate high-pressure storage vessel filled with a pressurant gas that is regulated down to the oxidizer tank pressure [20]. When compared to pump-fed variants, the pressure-fed systems are inherently simpler. Pressure-fed systems typically have larger inert masses since the propellant storage tank pressure must be higher than the combustion chamber pressure. Pump-fed systems can have relatively low storage pressure while maintaining higher chamber pressure, and thus have lower mass storage tanks due to thinner walls.

Pump-fed systems have typically used turbine-driven pumps, called turbopumps. In turbopump feed systems, the pump is driven by a turbine fed with a working fluid. Electric pump feed systems have a battery-powered electric motor in place of the classical turbine. The electric pump feed system has been considered since the 1990's for liquid bi-propellant systems and has been shown to be a higher-performing alternative to backfill pressure-fed configurations for hybrid systems [21]. Advances in battery technologies have made this feed system increasingly competitive [22].

1.2. Research objectives

The current study aims to find the minimum mass of two-stage-to-orbit hybrid rocket-propelled launch vehicles for a given payload mass, delivered to a 500 km polar orbit. The chosen orbit and altitude are commonly referenced to quantify launch vehicle capability polar launch provides no benefit to launch vehicle performance due to Earth's rotation. The minimization of dry mass is chosen, as the dry mass is assumed to be strongly linked to materials and manufacturing cost. The minimization of dry or gross mass in the absence of detailed cost data is a commonly-used strategy by other researchers [17,23]. The configurations of the optimal launch vehicles will be assessed through parametric studies.

1.3. Methodology

The design space of a hybrid rocket vehicle is divided into the various relevant sub-disciplines. The design variables, viewed as tunable parameters or constants, dictate the design of the launch vehicle as it propagates through the various sub-disciplines.

The propulsion system (propellants and propellant feed system) is considered first. After the propulsion system is designed, the propellant masses, propellant volumes, fuel geometries, propulsive performance (specific impulse), and converging-diverging nozzle geometries are defined. The vehicle structural configuration is then defined by vehicle elements (oxidizer tank, interstage, etc.). Structural and geometric inputs and constants (outer diameter, nosecone, etc.) further constrain the design. The aerodynamic modeling provides the aerodynamic forces, given vehicle geometry and the simulated environment, resulting in the ability to calculate the vehicle dynamics and loads. The structure is then iteratively sized, based on material yield stress and buckling failure modes using in-flight and on-pad structural loads. The trajectory is converged to the target altitude and speed by altering flight path conditions and trajectory control points, while minimizing propellant use. The described process results in a single vehicle design solution. Numerous vehicle designs are then evaluated and successive design iterations are altered in a global optimization.

2. Conceptual design models

The change in velocity (ΔV) is referenced throughout this study. The ΔV of a launch vehicle or spacecraft is a measure of the impulse per unit mass needed to perform a maneuver such as launching to orbit [24]. The ΔV of a launch vehicle can be reduced, under several assumptions, to the Tsiolkovsky rocket equation [25]:

$$\Delta V = I_{\mathrm{sp}} g_0 \ln\left(\frac{m_0}{m_{\mathrm{f}}}\right) \tag{1}$$

The derivation assumes an impulsive thrust and constant exhaust velocity. The derivation neglects aerodynamic drag, gravity, and steering losses. Under the array of assumptions, the ΔV is only dependent on the propulsive performance (specific impulse, I_{sp}) and structural efficiency (mass ratio, $\frac{m_0}{m_{\mathrm{f}}}$). The parameters m_0 and m_{f} are the initial and final vehicle masses, respectively. The terms in the rocket equation cannot be optimized one at a time, since the design space of a launch vehicle is highly coupled. A conceptual design framework is needed to propagate designs from design parameters and constants to the final configuration in order to generate, evaluate, and optimize designs.

2.1. Propulsion system

The initial propulsion system parameters include the hybrid rocket engines' nominal thrust ($\mathcal{T}_{\mathrm{nom}}$), and nominal total impulse

Table 1
Paraffin fuel attributes from Ref. [26].

Fuel parameter	Value
Regression constant - a	0.117×10^{-3}
Regression exponent - n	0.62

($I_{t,nom}$). The nominal quantities are defined for initial ambient back pressure (i.e. sea level or vacuum) and nominal combustion parameters. The burn time, t_b, can be equivalently used to replace either the nominal thrust or the nominal total impulse parameters, and is related to the other initial quantities by

$$t_b = \frac{I_{t,nom}}{\mathcal{T}_{nom}} \tag{2}$$

The propellants used in this study are paraffin wax with liquid oxygen (LOx). The axial-length-averaged fuel regression parameters of the propellant combinations are found in Ref. [26]. The regression coefficients are presented in Table 1. The a factor provides regression rates in m/s for flux values with units of $\frac{\text{kg}}{\text{m}^2\cdot\text{s}}$. The oxidizer and pressurant gas equilibrium thermodynamic fluid properties are found by use of equations of state accessed using the open source CoolProp package [27]. The LOx properties are determined assuming saturated liquid at an equilibrium temperature of 91 K.

The combustion chamber pressure (P_c) is less than the feed system outlet/injector upstream pressure (P_{up}), by a constant injector pressure drop (ΔP_{inj}). The oxidizer mass flow rate is assumed constant, which contributes to the validity of the constant injector pressure drop assumption. The maximum operating combustion chamber pressure will be defined from the upstream pressure, pressure drop across the injector, and the maximum pressure fluctuations from the mean (ΔP_{fluct}):

$$P_c \leq \frac{P_{up} - \Delta P_{inj}}{1 + \Delta P_{fluct}/P_c} \tag{3}$$

The deviation of chamber pressure fluctuations from the mean is 5%, which is in range of stable combustion [20]. In this study, for pressure-fed systems, the upstream pressure is the oxidizer tank pressure. For pump-fed engines, the upstream pressure is the pump discharge pressure. Constraints on injector pressure drop from liquid rocket injector design guidelines are used [28]:

$$\Delta P_{inj} \geq 0.5 P_c \tag{4}$$

The injector pressure drop is desired to be minimized such that the feed system pressure is the lowest for a given chamber pressure, and the inert mass is minimized. The optimal design occurs when the inequality constraints (Eqns. (3), (4)) are equal, and the chamber pressure can be completely defined as a fraction of the upstream pressure:

$$P_{up,\,min} = 1.55 P_c \tag{5}$$

The average oxidizer-to-fuel-ratio ($\overline{O/F}$) is defined as the ratio of oxidizer mass flow to fuel mass flow, averaged over the duration of the burn. The $\overline{O/F}$ may not equal the instantaneous oxidizer-to-fuel-ratio (O/F) as hybrid rockets can experience O/F shifts [29].

The NASA Chemical Equilibrium with Applications (CEA) [30] computer code is used to compute the propellant equilibrium combustion reaction properties. The reaction is assumed to be frozen at the throat and the combustion products are modeled as an imperfect gas. The frozen-at-the-throat assumption is conservative as it under-predicts performance, particularly for larger expansion nozzles [23,31]. The propulsion system models used in this study undergo verification and comparison to other models in Ref. [31]. The paraffin species from CEA's implementation is used, it is modeled as $C_{73}H_{124}$ with a heat of formation of -1860.6 MJ/mol. The paraffin attributes used produce similar specific impulse (within 2%) to others used in Ref. [32]. The calculation inputs are reaction O/F, combustion pressure (P_c), and nozzle area ratio ($\frac{A_e}{A_t}$). NASA CEA is used to calculate the ideal vacuum specific impulse ($I_{sp,vac,ideal}$), ideal characteristic velocity (c^*_{ideal}), and the nozzle exit pressure (P_e):

$$I_{sp,vac,ideal}, c^*_{ideal}, P_e = f(O/F, P_c, \frac{A_e}{A_t}, \text{fuel, oxidizer}) \tag{6}$$

where $f()$, describes a generic function. The c^*_{ideal} is adjusted by a combustion efficiency parameter (η_c) to provide c^*:

$$c^* = c^*_{ideal}\eta_c \tag{7}$$

The characteristic velocity is defined by the following [20]:

$$c^* \equiv \frac{P_c A_t}{\dot{m}_{total}} \tag{8}$$

The nozzle exit pressure variable (P_e) is introduced and is constrained to be greater than 40% of ambient pressure (P_{amb}) as dictated by the Summerfield flow separation criteria [33]:

$$P_e > 0.4 P_{amb} \tag{9}$$

NASA CEA provides the ideal vacuum specific impulse, or ideal perfectly expanded specific impulse. To obtain the general ambient specific impulse, the following equation is used:

$$I_{sp,amb} = \eta_c\left(I_{sp,vac,ideal} - \frac{c^*_{ideal}}{g_0}\frac{P_{amb}}{P_c}\frac{A_e}{A_t}\right) \tag{10}$$

The value of reference gravity (g_0) used in this study is the value used by NASA CEA: 9.806635 m/s^2.

The instantaneous thrust ($\mathcal{T}$) is calculated from:

$$\mathcal{T} = \dot{m}_{total} I_{sp,amb} g_0 \tag{11}$$

The average total mass flow rate ($\overline{\dot{m}}_{total}$) is determined by using Eqn. (11) and replacing $\mathcal{T}$ with nominal thrust ($\mathcal{T}_{nom}$) and using $I_{sp,amb}$ at ignition conditions, with $\overline{O/F}$ and nominal P_c.

2.1.1. First stage nozzle formulation

For first stage nozzles, the above formulation is used with nozzle exit pressure provided as an input constrained by flow separation criteria. Equation (10) is solved iteratively to find the area ratio for a given ratio of exit pressure to chamber pressure. The required total mass flow rate for the first stage is calculated by modifying Eqn. (11) and using the thrust and specific impulse at nominal conditions and initial back pressure. The throat area is determined by modifying Eqn. (8) with nominal P_c, average mass flow ($\overline{\dot{m}}_{total}$), and c^* (Eqn. (7)).

2.1.2. Upper stage (vacuum) nozzle formulation

For upper stage nozzles, the ambient pressure is usually small and the exit area is desired to be large. In this study, the maximum exit diameter is constrained by the vehicle body inside diameter rather than the exit pressure. The nozzle exit diameter (D_{exit}) is equal to the housing diameter ($ID_{interstage}$) minus an introduced non-dimensional wall clearance parameter (δ_{nozzle}):

$$D_{exit} = (1 - 2\delta_{nozzle}) ID_{interstage} \tag{12}$$

The δ_{nozzle} parameter is the radial length between the nozzle bell and the inner diameter of the interstage, divided by the inner diameter of the interstage. This non-dimensional wall clearance parameter has a lower bound to ensure feasible designs are generated with complete design freedom to make any scale nozzle, within the bounds of the interstage:

$$0.05 \leq \delta_{\text{nozzle}} < 0.5 \tag{13}$$

The upper stage nozzle throat area is determined iteratively by minimizing the error of the input vacuum nominal thrust parameter ($\mathcal{T}_{\text{nom}}$) and the thrust output from Eqn. (11). The $\overline{\dot{m}}_{\text{total}}$, used in Eqn. (11), in lieu of $\dot{m}_{\text{total}}$, is calculated from Eqn. (8) assuming nominal P_c. The initial $I_{\text{sp,amb}}$ is calculated assuming zero back pressure and nominal P_c and $\overline{O/F}$.

2.1.3. Propellant masses and geometry

The average oxidizer and fuel mass flow rates are determined using $\overline{\dot{m}}_{\text{total}}$ and $\overline{O/F}$:

$$\overline{\dot{m}}_{\text{ox}} = \overline{\dot{m}}_{\text{total}} - \overline{\dot{m}}_{\text{f}} = \frac{\overline{\dot{m}}_{\text{total}}}{1 + (\overline{O/F})^{-1}} \tag{14}$$

The variables $\dot{m}_{\text{ox}}$ and $\dot{m}_{\text{f}}$ are the oxidizer and fuel mass flow rates, respectively. The individual propellant masses are determined by the average mass flow rates and burn time:

$$m_{\text{i}} = \overline{\dot{m}}_{\text{i}} \cdot t_{\text{b}} \tag{15}$$

where the subscript i denotes an arbitrary propellant (fuel or oxidizer).

A single-port, circular-cylinder fuel grain geometry is used in this study. The spatially-averaged fuel surface regression rate ($\dot{r}_{\text{f}}$) power law relation [26] is used:

$$\dot{r}_{\text{f}} = \text{a}G_{\text{ox}}^{\text{n}} \tag{16}$$

The parameters a and n are empirically determined constants that are unique to each fuel and oxidizer combination. The regression constants used in this study are presented in Table 1. G_{ox} is the oxidizer mass flux defined by

$$G_{\text{ox}} = \frac{\dot{m}_{\text{ox}}}{A_{\text{p}}} = \frac{\dot{m}_{\text{ox}}}{\frac{\pi}{4}(ID_{\text{f}})^2} \tag{17}$$

and A_{p} is the internal port area. The parameter ID_{f} is the inner diameter of the fuel grain.

The fuel internal port diameter is minimized to improve volumetric packing efficiency. The minimum port diameter is constrained by the maximum allowable mass flux of the oxidizer. Oxidizer mass flux values up to 500 $\frac{\text{kg}}{\text{m}^2\cdot\text{s}}$ have been demonstrated in operational hybrid rocket motors [34]. Additionally, a Mach number constraint is added to the fuel initial port diameter to ensure it is not a choke point and has equal or greater area to the exhaust nozzle.

An axial-length-averaged, time-resolved numerical fuel regression simulation is used to solve for the fuel mass flow rate. Given the oxidizer mass flow rate history, the fuel mass flow rate is determined by fuel surface regression rate (Eqn. (16)). Given that the regression is assumed independent of fuel axial length, the required fuel outer diameter is determined. The numerical simulation produces an O/F history based on the ratio of mass flow rate histories. The numerical time-averaged O/F is then converged to the input parameter $\overline{O/F}$ by altering the length of the fuel grain iteratively.

A pre-combustion chamber section (a void volume upstream of the fuel grain within the combustion chamber) is added and is assumed to be equal to the fuel outer diameter in length. A post-combustion chamber section (a void volume downstream of the fuel grain) is also included and assumed to be 1.5 times the fuel outer diameter in length. These values were adapted from motor geometry of an operational hybrid rocket engine [35].

The number of engines for a first stage is selected to be seven. The optimal packing efficiency occurs with one motor [36], but seven engines result in a local optimal packing efficiency if more than one engine is needed. Multiple engines are desired to reduce the individual engine size and reduce manufacturing and testing complexity. The number of upper stage engines is set to one, which allows a relatively large, radiatively-cooled engine nozzle diverging sections. In this study, the thrust, average oxidizer flow, and fuel mass is split equally between all engines for each separate stage.

2.1.4. Time-varying propulsion system properties

The propulsion system is initialized assuming nominal conditions; however, the time-resolved characteristics must be determined. In general, the O/F is expected to shift, despite assuming the oxidizer flow rate is constant for this study. The O/F shift will change P_c simultaneously since overall the mass flow rate changes. Since the change in chamber pressure is dependent on the change in mass flow rate and c^*, and c^* is, in this study, a function of chamber pressure, an iterative process is required to determine the chamber pressure. The difference in characteristic velocity determined from Eqn. (8) and Eqn. (6) (multiplied by η_c) is minimized. The change in O/F and P_c is used to calculate the c^* and $I_{\text{sp,vac}}$ histories used for each instant of the burn duration. The $I_{\text{sp,amb}}$ and $\mathcal{T}$ are extracted from Eqn. (10) and (11), respectively, at each required instant.

2.1.5. Feed system design and scaling parameters

The feed systems considered for the current study are gas pressure-fed (backfill) and electric pump-fed. The pump-fed configurations still require a backfill pressurant, but typically at lower pressures to only satisfy net positive suction pressure (NPSP) requirements.

The pressurant mass is determined assuming an adiabatic ideal gas expansion from the pressurant tank volume to the total oxidizer tank volume ($\forall_{\text{ox}}$) and oxidizer pressure (P_{ox}) [23]:

$$m_{\text{pg}} = c_{\text{pg}} \frac{P_{\text{ox}} \forall_{\text{ox}}}{R_{\text{pg}} T_{\text{i,pg}}} \left(\frac{\gamma_{\text{pg}}}{1 - \frac{P_{\text{ox}}}{P_{\text{i,pg}}}} \right) \tag{18}$$

The resulting mass is multiplied by a factor, c_{pg}, to provide a margin. A value of 1.2 is used for c_{pg} in this study as was used by [23]. The initial temperature ($T_{\text{i,pg}}$) and pressure ($P_{\text{i,pg}}$) of the pressurant gas are input design parameters. The pressurant gas properties, R_{pg} and γ_{pg}, are the specific gas constant and ratio of specific heats, respectively.

2.1.6. Electric pump feed system

The electric pump feed system requires a pressurant system, electric battery, and pump system. The battery and pump system are sized given the power and energy requirements. The battery and pump scaling laws are adapted from Ref. [21] The pump instantaneous power is expressed by

$$\mathcal{P}_{\text{ep}} = \frac{\dot{m}_{\text{ox}} \Delta P_{\text{ep}}}{\rho_{\text{ox}} \eta_{\text{ep}}} \tag{19}$$

where $\mathcal{P}_{\text{ep}}$ is the pump power, η_{ep} is the electric pump efficiency, and ΔP_{ep} is the difference between the inlet and outlet oxidizer stream pressure. The total energy is the temporal integral of the power over the burn duration:

$$E_{\text{ep}} = \int_0^{t_{\text{b}}} \mathcal{P}_{\text{ep}} \text{d}t \tag{20}$$

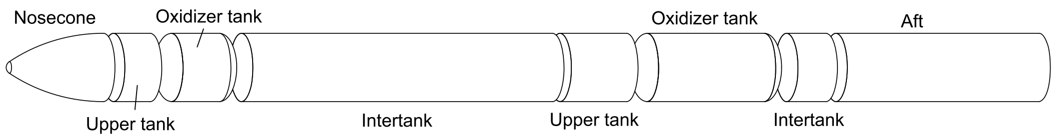

Fig. 1. Conceptual layout of rocket elements.

Table 2
Electric pump constants from Refs. [21,37,38].

Electric pump attribute	Value
Pump assembly power density (δ_{ep})	3.9 kW/kg
Battery power density (δ_{bp})	3.0 kW/kg
Battery energy density (δ_{be})	324.0 kJ/kg
Pump efficiency (η_{ep})	66%
Required NPSP	350 kPa
Battery mass density	1000 kg/m^3
Pump assembly mass density	500 kg/m^3

The battery mass (m_{b}) is determined from an empirical battery energy density (δ_{be}) and battery power density (δ_{bp}). The battery mass is determined from the maximum of the energy-limited or power-limited cases:

$$m_b = \max(\frac{\mathcal{P}_{\mathrm{ep,max}}}{\delta_{\mathrm{bp}}}, \frac{E_{\mathrm{ep}}}{\delta_{\mathrm{be}}}) \tag{21}$$

The pump mass (m_{ep}) is calculated using an assumed constant power density (δ_{ep}):

$$m_{ep} = \frac{\mathcal{P}_{\mathrm{ep,max}}}{\delta_{\mathrm{ep}}} \tag{22}$$

The pump mass includes the pump housing, electric motor, and inverter which can all be taken as a function of the power requirement [37]. Electric pump values used in this study are adapted from Ref. [21,37,38] and are presented in Table 2.

The NPSP is the absolute pressure at the inlet eye of the centrifugal pump (Eqn. (23)). The available NPSP ($\mathrm{NPSP_A}$) is a function of pressurant gas pressure (P_{pg}), static pressure ($P_{\mathrm{st,ox}}$) in the oxidizer tank, and the vapor pressure of the oxidizer ($P_{\mathrm{vap,ox}}$):

$$\mathrm{NPSP_A} = P_{\mathrm{pg}} + P_{\mathrm{st,ox}} - P_{\mathrm{vap,ox}} \tag{23}$$

The static pressure ($P_{\mathrm{st,ox}}$) is calculated from oxidizer density (ρ_{ox}), apparent acceleration ($|\overrightarrow{a}|$, which includes gravitational acceleration), and oxidizer liquid level height (h_{ox}):

$$P_{\mathrm{st,ox}} = \rho_{\mathrm{ox}}|\overrightarrow{a}|h_{\mathrm{ox}} \tag{24}$$

The $\mathrm{NPSP_A}$ is required to be high enough to avoid cavitation during the operation of the electric pump in order to prevent significant mechanical wear to the impeller.

$$\mathrm{NPSP_A} \geq \mathrm{NPSP_R} \tag{25}$$

The required NPSP ($\mathrm{NPSP_R}$) is dictated by the detailed design of the impeller. A constant value is assumed based on existing literature values [39].

2.2. Overall structure and geometry

To limit the design space, the vehicle outer diameter (D_{O}) is constant along the length with an ogive nosecone. The D_{O} is greater than the outer diameter of the first stage combustion chambers and nozzles, and second stage combustion chamber. The rocket is composed of elements with load-bearing shells that experience and transmit aerodynamic, inertial, internal pressure, and thrust loads. The vehicles elements are depicted in Fig. 1. Each element's structural section is modeled as a thin cylinder with constant wall thickness. Each element has properties of mass, mass-moment of inertia, and center of mass. The internal volume is calculated from the required mass and density of the internal components. All component masses are assumed to be distributed uniformly within an element. The nosecone is assumed to be a hollow cone for the calculation of center of gravity and moment of inertia. The payload point mass is assumed to be located at the nosecone center of mass.

2.2.1. Nosecone

The nosecone fineness ratio (nosecone length divided by D_{O}) is 1.5. The bluntness ratio (nosecone radius divided by diameter) is 0.05. The nosecone mass is comprised of the payload mass, and the fairing (nosecone shell) mass. In this study, the fairing separation is performed at an altitude of 110 km where heat flux and dynamic pressure loads are low.

2.2.2. Upper tank

The upper tank components, consist of avionics, pressure transducers, valves, and a pressurant tank. The pressurant tank is assumed spherical or cylindrical if necessary. The avionics and pressure transducer masses are assumed to be 3.5 kg and 0.5 kg, respectively, per stage. The constant component mass values are from an operational sounding rocket [35], and area assumed constant across vehicle scales. Each upper tank section has two valves sized for oxidizer tank venting and pressurant gas injection. The valve mass is calculated based on the semi-empirical relation [40].

2.2.3. Oxidizer tank

The oxidizer tank is modeled as a cylindrical tank with hemispherical ends. The structure geometry, material density, fluid volumes, and fluid densities are used to calculate mass, center of gravity, and mass-moment of inertia.

2.2.4. Intertank

The intertank houses the valves, feed lines, electric pumps, batteries, and thrust structure. The thrust structure spans the length of the intertank. The cross-sectional area of the thrust structure is sized per engine, assuming simple axial compression. The feed system line flow area is limited by a design velocity and oxidizer volumetric flow per engine. The flow area, line pressure, feed line material, and intertank length provide the feed line mass using thin-walled pressure vessel theory which is detailed in Sec. 2.5.1. When the intertank is part of the upper stage, the mass and length of the combustion chambers and nozzles are added. The structural cylindrical shell of the interstage remains with the first stage upon staging.

2.2.5. Aft

The aft section consists of combustion chambers, fuel, nozzles, and a structural shell that extends from the end of the intertank to the inlet of the nozzles. The nozzle geometry is generated using the bell (parabolic) formulation outlined in Ref. [20].

The nozzle mass is determined from the addition of a nozzle mass estimating relation (MER) and a stress-based method. The nozzle MER is generated from empirical data by Ref. [41]. The nozzle MER accounts for thrust vector control components. The additional mass from the stress-based method accounts for the larger

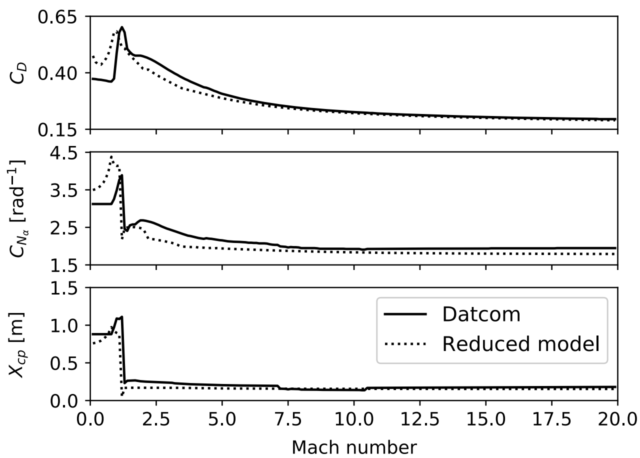

Fig. 2. Aerodynamic quantities of Datcom and reduced model.

divergent nozzle bells needed for higher expansion ratio nozzles, since most of the data used to generate the nozzle MER is from first stages/boosters.

2.3. Aerodynamics

A component-buildup aerodynamic method is used in the current study due to its computational speed and the relatively simple geometry of the simulated vehicles. The investigated aerodynamic coefficients include: drag coefficient (C_D), the normal force coefficient derivative with angle of attack (C_{N_α}), and the center of pressure (X_{cp}). All aerodynamic coefficients are calculated at zero angle of attack (α) since small angles of attack are assumed in the study.

The nosecone aerodynamic quantities are calculated using the methods contained in 1997 Missile Datcom [42]. In order to increase computational speed, the cylindrical body aerodynamic methods from Datcom are not used. The cylindrical body aerodynamic methods are from Ref. [43]. The base drag is given by the relation from Ref. [44]. The lightweight modeling used in this study was compared and verified with 1997 Missile Datcom for the coefficients of interest. The model agreement for a vehicle with an aspect ratio of 20 is presented in Fig. 2. The aerodynamic methods gradually become identical as the aspect ratio is further reduced. Additional model formulation details can be found in Ref. [31].

2.4. Flight simulation

To model the translational motion, the vehicle is modeled as a particle, ignoring the size and mass distribution. Modeling the translational motion is sufficient for the preliminary design and analysis of vehicle trajectories [23]. In the vehicle reference frame, a Cartesian, 3 degree of freedom (DOF) formulation is used for determining flight loads and angular acceleration. The 3 DOF frame is assumed to be coincident with the orbital plane.

2.4.1. Equations of motion

The equations of motion, adapted from Ref. [45], are presented below.

$$\dot{r} = v \sin\gamma \tag{26}$$

$$\dot{\xi} = \frac{v\cos\gamma\cos\zeta}{r\cos\phi} \tag{27}$$

$$\dot{\phi} = \frac{v\cos\gamma\sin\zeta}{r} \tag{28}$$

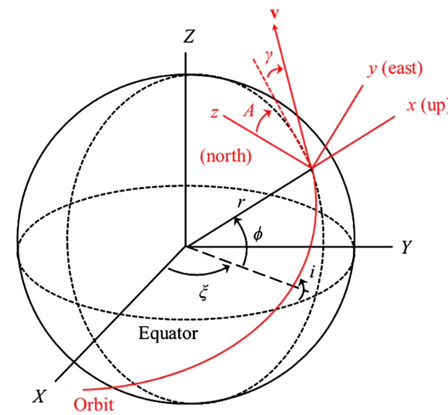

Fig. 3. Earth-centered and local horizontal reference frames, adapted from Ref. [45].

$$\dot{\gamma} = \frac{\mathcal{T}\sin\beta_T + N}{mv} + (\frac{v}{r} - \frac{g(r)}{v})\cos\gamma + \cos\phi[2\omega_E\cos\zeta + \frac{\omega_E^2 r}{v}(\cos\phi\cos\gamma + \sin\phi\sin\gamma\sin\zeta)] \tag{29}$$

$$\dot{v} = \frac{\mathcal{T}\cos\beta_T - D}{m} - g(r)\sin\gamma + \omega_E^2 r\cos\phi(\cos\phi\sin\gamma - \sin\phi\cos\gamma\sin\zeta) \tag{30}$$

$$\dot{\zeta} = -\frac{v}{r}\tan\phi\cos\gamma\cos\zeta + 2\omega_E(\cos\phi\tan\gamma\sin\zeta - \sin\phi) - \frac{\omega_E^2 r}{v\cos\gamma}\sin\phi\cos\phi\cos\zeta \tag{31}$$

For the above equations, note the following symbols, also shown in Fig. 3:

- $\mathcal{T}$ - thrust $[N]$
- D - drag $[N]$
- N - normal force (lift) $[N]$
- g - Earth's local gravitational acceleration $[\mathrm{m/s^2}]$
- ω_E - Earth's angular velocity $[rad/s]$
- β_T - angle of thrust vector wrt. velocity vector $[rad]$
- ϕ - latitude $[rad]$
- ξ - longitude $[rad]$
- A - azimuth $[rad]$
- ζ - heading angle $(= \frac{\pi}{2} - A)$ $[rad]$
- γ - flight path angle $[rad]$

The above equations (Eqn. (26)–(31)) describe the motion of a body (local horizontal reference frame) moving in a rotating reference frame (Fig. 3). This formulation describes the motion in 3D space, but neglects vehicle roll about the longitudinal axis.

The atmospheric properties are modeled using 1976 US Standard Atmosphere [46]. The equations of motion, integration scheme, gravitational acceleration, and atmospheric properties are verified in Ref. [31].

2.4.2. Scalar wind modeling

The scalar wind envelope, synthetic wind profiles, and gust magnitudes are provided using methodology outlined in Ref. [47]. The 99% probability scalar wind speed envelope is used from Table 2-53 in Ref. [47]. The scalar envelope defines the maximum speed that a wind profile can have at any given altitude above sea level (ASL). The synthetic profile describes a possible scalar wind profile for a single flight. The steady-state envelope and synthetic wind profile are illustrated in Fig. 4.

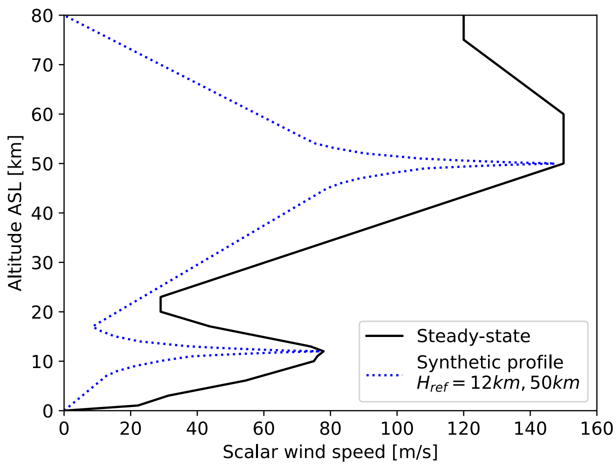

Fig. 4. Scalar wind steady-state envelope with synthetic wind profile.

Each flight simulation uses the same synthetic wind profile. The synthetic profile has maximum wind values that are defined at two reference altitudes. Reference altitudes of 12 km and 50 km are used in this study because they lead to the largest winds and loads. The reference altitudes have corresponding "build-up" and "back-off" profiles which describe how wind speed changes with altitude. These profiles are provided from Ref. [47]. In addition to the synthetic wind profile speeds, the wind gust is included at all altitudes and modeled as a discrete 9 m/s gust [47]. The gust is modeled to always increases angle of attack (worst-case).

2.4.3. Trajectory

For the present study, the trajectory consists of distinct phases: vertical take-off, pitch over, gravity turn, bilinear tangent steering law [23], and a second-stage coast and re-light to circularize. Once a simulated launch vehicle reaches 100 m (clears the launch tower), the vehicle starts a pitch-over maneuver to a specified pitch-over angle (γ_{po}). The pitch-over maneuver occurs before aerodynamic forces are large, and with a constant pitch rate of 0.005 rad/s. The vehicle then undergoes a gravity turn, where Earth's gravity is the only force altering the trajectory. Once staging occurs, after a 1 s coast time, the upper stage uses the bilinear tangent steering law. The bilinear tangent steering law provides the flight path angle (γ) as a function of normalized time (τ_b) into the burn:

$$\gamma = \begin{cases} \tan^{-1}\left(\dfrac{c^{\chi}\tan\gamma_i + (\tan\gamma_f - c^{\chi}\tan\gamma_i)\frac{\tau_b}{b}}{c^{\chi} + (1-c^{\chi})\frac{\tau_b}{b}}\right) & 0 \le \tau_b \le b \\ \gamma_f & \tau_b > b \end{cases} \tag{32}$$

where,

γ - flight path angle [rad]
γ_i - initial flight path angle[rad]
γ_f - final flight path angle (typically 0) [rad]
c - trajectory shape control parameter, $10 \le c \le 100$
χ - trajectory exponent parameter, $-1.0 \le \chi \le 1.0$
τ_b - normalized burn time, $0.0 \le \tau_b \le 1.0$
b - burn time modification parameter, $0 < b \le 1.0$

The parameters c and χ define the shape of the trajectory. The parameter b is not part of the classic bilinear tangent steering law. Parameter b is introduced to modify τ_b to create flight profiles which become tangent to the Earth's surface sooner than the classic method. During the tangent flight section, the ability to truncate the burn provides an easier means of obtaining a desired orbit compared to altering the entire trajectory with c and χ. The initial flight path angle of the second stage (γ_i) is equal to the final flight path angle of the first stage. The final flight path angle of the second stage (γ_f) is 0 rad (tangent to the Earth's surface). During the second stage flight, the flight path angle in the equations of motion is prescribed by the value of Eqn. (32). The assignment of Eqn. (32) to the flight path heading of the second stage neglects the steering losses, which have been observed to be small for upper stages given the lack of aerodynamic forces [23].

During the second stage burn, before the initial second-engine cut-off (SECO-1), the orbital elements from the state vector. The six arguments of the position and velocity vector are translated into six Keplerian elements that define the orbit path in space [24]. When the calculated orbit apogee reaches the target apogee, the burn is truncated. The initial orbit at SECO-1 is elliptical, whereas a circularization burn makes the final orbit circular. The numerical flight simulation does not persist after SECO-1 to increase computational speed. The required ΔV for circularization is calculated using the rocket equation (Eqn. (1)) with the average remaining I_{sp}. The rocket equation's assumptions are valid for the circularization burn. An impulsive-thrust orbit transfer assumption is made. The impulsive-thrust assumption is valid for chemical rockets since the burn times are small fractions of the total orbit period [24].

The trajectory design is multi-objective which includes solving a boundary-value problem (minimizing error from the desired end altitude and speed) and minimizing propellant usage. The trajectory is optimized by altering the parameters: γ_{po}, c, χ, and b. The cost function used to optimize upper stage trajectories is the error of the SECO-1 altitude from the target, subtract the ΔV of the remaining propellant residual on board after circularization. The ΔV is calculated using Eqn. (1), with the remaining average vacuum specific impulse and remaining propellant and inert masses. The algorithm used to generate upper stage trajectories is the L-BFGS-B algorithm in the Scipy Optimize module [48].

2.5. Structural design

2.5.1. Pressure vessels

The conceptual design of the oxidizer tanks, pressurant tanks, and the combustion chamber is accomplished with the use of thin-walled pressure vessel theory [49]. In this study, the oxidizer tank outer diameter is equal to the vehicle outer diameter. The oxidizer tank has a secondary function as the airframe structure which avoids redundancy and increases mass-efficiency. The pressurant tank is modeled as a spherical vessel. If a spherical vessel cannot fit within the body diameter, a cylindrical pressure vessel with hemispherical end caps is substituted. The cylindrical pressure vessel hoop and axial stresses are presented by

$$\sigma_{hoop} = \frac{Pr}{t} \tag{33}$$

$$\sigma_{axial} = \frac{Pr}{2t} \tag{34}$$

where, P is the internal pressure, r is the radius (half the vehicle outer diameter), and t is the wall thickness. The two stresses above represent principal stresses in the material. With the knowledge of the material and its failure mode, a failure criteria can be chosen. A common failure criterion for ductile materials is the von Mises failure criteria [49]. The von Mises failure criteria for a thin-walled pressure vessel cylinder is presented by

$$\sigma_y = \frac{\sqrt{3}}{2}\frac{Pr}{t} \tag{35}$$

The tank-mass-to-fluid-mass ratio is found to not be a function of geometry, but rather a function of tank material strength

(σ_{allow}), material density (ρ_{material}), internal pressure, and fluid density (ρ_{fluid}). The tank mass scaling ratio is provided below:

$$\frac{m_{\text{fluid}}}{m_{\text{tank}}} = \frac{\rho_{\text{fluid}}\sigma_{\text{allow}}}{\sqrt{3}\rho_{\text{material}}P} \tag{36}$$

A detailed formulation is presented in Ref [31].

Flat plates, and hemispheres were investigated for the end caps of the pressure vessels. Flat end caps are used for the combustion chambers, whereas hemisphere caps are used for the oxidizer tanks. For hemispherical end caps or spherical vessels, both principal stresses are equal and the von Mises criteria becomes the following:

$$\sigma_y = \frac{Pr}{2t} \tag{37}$$

For flat plates, with clamped edge boundary conditions, the maximum stress occurs in the radial direction at the edge of the plate [49]. This stress is equated to tensile yield for a failure criteria:

$$\sigma_y = \frac{3P}{4}\left(\frac{r}{t}\right)^2 \tag{38}$$

2.5.2. Combined loading

In general, for a cylindrical rocket element, loading can occur in the forms of: axial force, bending moment, and internal pressure. The body forces (vehicle weight and inertial loads), external aerodynamic forces, and thrust provide a net axial force (F_{ax}). The axial stress is determined with the net axial force: $\sigma_{\text{ax}} = \frac{F_{\text{ax}}}{A}$. The parameter A is the cross-sectional area of a thin cylindrical section. Bending loads (M) arise from lateral forces due to the vehicle's response to non-zero angle of attack, lateral components of body forces, and lateral components of thrust. The maximum and the minimum bending loads occur on opposite sides of the neutral axis: $\sigma_b = \pm\frac{Mr}{I}$. The parameter I is the sectional area moment of inertia, and the values are from Ref. [49]. Stresses from internal pressure are expressed by Eqn. (33) and (34). The von Mises stress is calculated using the principal stress values for the axial and circumferential dimensions. The first principal stress is the sum of the axial, bending, and axial internal pressure stress:

$$\sigma_1 = \frac{F_{\text{ax}}}{A} \pm \frac{Mr}{I} + \frac{Pr}{2t} \tag{39}$$

The second principal stress is the hoop stress due to internal pressure:

$$\sigma_2 = \frac{Pr}{t} \tag{40}$$

In addition to von Mises failure criterion, thin cylinder buckling criterion is used from Ref. [50]. The buckling criterion is analyzed considering axial compression, bending, and internal pressure. For further details on the implementation of this buckling criterion see Ref. [31].

2.5.3. Flight loads

The body and aerodynamic forces are combined into resultants at the center of mass and pressure, respectively, for each rocket element. The forces are further decomposed into axial forces, lateral shear forces, and bending moments at the rocket element interfaces.

At each instant in flight, the loads are calculated assuming the current wind is given by the steady-state 99% envelope in addition to a discrete sharp-edged gust. The additional gust creates a dynamic response of the vehicle. During the dynamic response, angular acceleration and lateral acceleration contribute to inertial loads and relief; however, the aerodynamic plunge and pitch damping are neglected. The lateral thrust from the first stage engines trims the vehicle against the aerodynamic forces from the steady-state wind to create a zero moment about the vehicle center of mass. The act of trimming can contribute a large bending stress on the vehicle.

The axial force, shear force, and bending moment equations are derived with the nosecone tip as the origin. Bending, shear, and axial loads are zero at the free ends unless a load (e.g. thrust) is prescribed there. The successive loads are determined by spatial marching along the vehicle. The detailed derivation is presented in Ref. [31]

2.5.4. Special load cases

In addition to the in-flight loads, a few special load cases are added to size various elements:

- Loss of pressure on-pad
  The load case of a depressurized vehicle on the pad is analyzed. All masses onboard are the starting values (full propellant loading, worst case), but all propellant tanks have zero gauge pressure.
- Aft engine section skirt axial compression
  This load case checks the aft section around the main engines while bearing the total vehicle weight in compression, statically on the launch pad.
- Upper stage flight loads
  The lateral loads on the second stage are neglected during the flight of the second stage. However, two important quantities from the upper stage flight are still calculated and included. The maximum axial loads and maximum fluid static pressure are calculated for upper stages assuming zero back pressure for the propulsion system, and aligned velocity and gravity vectors (worst case).

2.5.5. Modal analysis

The displacement/stiffness constraints are another important aspect of the launch vehicle's structural design. The stiffness requirements are based on a modal (frequency domain) analysis of a finite-beam-element model of a rocket. The rocket model, as previously discussed, is divided into discrete elements with mass and section properties. The stiffness constraints are the lowest allowable bending and axial mode fundamental vibrational frequencies. The fundamental modes increase in frequency value as overall rigidity is increased. The values of lower bounds are 25 Hz and 10 Hz for axial and bending modes respectively, and are taken from launch vehicle design criteria and based on previous existing launchers from Ref. [51]. The rigidity constraints are important for maintaining control stability and avoiding any aeroelastic instabilities, that could lead to high fatigue loading or divergence.

In the modal analysis formulation, each rocket element is modeled as a cylinder with uniform mass distribution. The mass and stiffness matrices are determined for each element based on Bernoulli-Euler beam theory [52]. The global stiffness and mass matrices are constructed from all local element matrices. The non-trivial solutions of the eigenequation for the finite element model of the rocket are solved, and fundamental modes are quantified, when the following is satisfied:

$$det([\mathbf{K}] - \omega^{\mathbf{2}}[\mathbf{M}]) = \mathbf{0} \tag{41}$$

where, $[\mathbf{K}]$ is the global stiffness matrix and $[\mathbf{M}]$ is the global mass matrix.

The axial and bending modes are considered separately and the two eigenvalue problems are separately solved, one with longitudinal stiffness matrix and one with lateral stiffness matrix. The decoupled methodology is used such that the calculated modes

Table 3
Material properties of rocket structures.

Name	ρ [kg/m^3]	σ_y [MPa]	E [GPa]
2024-T6	2780	345	73.1
Ti-6Al-4V	4429	880	96.9
Ir/Re	21000	300	N/A

could be easily compared to the existing criteria. A 0 Hz mode is present for the longitudinal and lateral directions, implying rigid body motion. Rigid-body motion is expected due to the free-free boundary conditions imposed. The lowest non-zero values are used as the first fundamental modes of vibration. Verification and detailed formulation can be found in Ref. [31].

2.5.6. Materials

The majority of the structural elements used a high strength-to-weight aluminum alloy: 2024-T6. The 2024-T6 alloy is used in aerospace applications, material properties for 2024-T6 are taken from Ref. [53]. Pressurant tanks and combustion chambers are modeled to have Ti-6Al-4V material properties. Material properties for Ti-6Al-4V are taken from Ref. [54]. While carbon-fiber composite materials could decrease overall mass, minimizing the fabrication costs and complexity for small-scale launch vehicles was prioritized. The nozzle diverging structure was modeled as iridium-coated rhenium (Ir/Re) [55] which has favorable thermal-structural properties for radiatively-cooled rocket nozzles and had readily available material property data. All material property data, from the mentioned references, is organized in Table 3. All safety factors had a value of 1.5 apart from combustion chamber and nozzle structures, which were set to 2.0. All safety factors are applied to critical stress values defined by the failure criteria.

2.5.7. Structural analysis program sub-routine

During the simulations, the structure wall thicknesses are sized incorporating the stresses incurred during flight. However, since the flight performance is affected by the dry mass, determining the structure thicknesses is iterative. During the structural analysis, the buckling/yield safety margins are first converged to the target margins by altering all section wall thicknesses using a linear least-squares error minimization. The linear least squares error minimization used a Trust Region Reflective algorithm which was analyzed as the most robust/fastest method and is available in the Scipy Optimize Python module *least_squares* function [48].

After the buckling/yield requirements are met, the fundamental mode (rigidity) constraints are assessed. If the fundamental vibration mode constraints are violated, the thickness of each section are modified in another local-level optimization. The thickness lower bounds for the modal analysis are taken from the output of the strength-based structural analysis. The local level optimization minimizes structure mass while adjusting all section thicknesses to meet the desired rigidity criteria. Further program description can be found in Ref. [31].

2.6. Global optimization methodology

Particle swarm optimization (PSO) was selected since other authors reported the success of PSO alongside genetic algorithms (GAs), for launch vehicle conceptual design optimization [17,23,56,57].

The Python module Pyswarms [58] was used. Pyswarms has a user-friendly, tested, and maintained implementation of the algorithm. Evolutionary-based algorithms were of top consideration compared to the derivative-based methods used in inner-loops since the design space may not be sufficiently smooth and present local optimal solutions.

In PSO, a swarm of particles is placed randomly over n-dimensional space, where n is the number of design dimensions. The particles start with random initial velocity. The velocity of a single particle is then changed depending on what the individual perceives as the best locally, and what the consensus of the nearest particles perceives as the best. In this study, each particle is aware of the entire population's consensus of the perceived optimal and as such is "fully informed". The individual and social "forces" are weighted by constants c_1 and c_2 respectively. The particle may have inertia from mass parameter w to resist instantaneous direction changes. The position (x) and velocity (v) of the particles in the design space is governed by

$$x_i(t+1) = x_i(t) + v_i(t+1) \tag{42}$$

$$\begin{aligned} v_{ij}(t+1) = w v_{ij}(t) + c_1 r_{1j}(t)[y_{ij}(t) - x_{ij}(t)] \\ + c_2 r_{2j}(t)[\hat{y}_j(t) - x_{ij}(t)] \end{aligned} \tag{43}$$

The global optimization is placed at the output of the entire conceptual design process (Fig. 5). In Fig. 5 the schematic of the program structure is shown, including iteration loops and general program flow. The structures and trajectory are optimized/converged at the local level. The global cost function minimizes dry mass while enforcing design feasibility constraints (positive propellant residuals, staging height constraints, and initial altitude constraints). Since the cost function (Eqn. (45)) maintains some feasibility of the design, the process is part Inter-Disciplinary Feasible (IDF) [59]. Further details can be found in Ref. [31] regarding the program layout with the aforementioned structural analysis subroutine.

The inert mass payload fraction is maximized. It is assumed the largest cost of the vehicle will be in material costs of the structure rather than propellant.
The inert mass payload fraction, ϵ, is defined by

$$\epsilon = \frac{m_{\text{payload}}}{m_{\text{dry}}} \tag{44}$$

where m_{dry} is the total, initial vehicle dry mass (no propellants).
The global cost function for the PSO is defined by:

$$\text{Cost} = C_0 + C_1 \Delta V_{\text{res}} + C_2 |\Delta_{\text{H,staging}}| + C_3 |\Delta_{\text{H,SECO,1}}| \tag{45}$$

The weights (C_0, C_1, C_2, C_3) of the terms in the cost function are scaled with inert mass fraction (ϵ) and are determined using the following:

$$C_0 = \frac{100}{\epsilon} \tag{46}$$

$$C_1 = \begin{cases} \frac{100}{\epsilon}, & \Delta V_{\text{res}} < 0 \\ -0.1, & \Delta V_{\text{res}} \geq 0 \end{cases} \tag{47}$$

$$C_2 = \begin{cases} 100\epsilon, & \Delta_{\text{H,staging}} < 0 \\ 0, & \Delta_{\text{H,staging}} \geq 0 \end{cases} \tag{48}$$

$$\Delta_{\text{H,staging}} = H_{\text{staging}} - H_{\text{staging,minimum}} \tag{49}$$

$$C_3 = \begin{cases} \frac{\epsilon}{100}(\frac{|\Delta_{\text{H,SECO,1}}|}{15000} - 1), & |\Delta_{\text{H,SECO,1}}| \geq 15000 \\ 0, & |\Delta_{\text{H,SECO,1}}| < 15000 \end{cases} \tag{50}$$

$$\Delta_{\text{H,SECO,1}} = H_{\text{SECO,1}} - H_{\text{SECO,1,target}} \tag{51}$$

The weighting functions (Eqns. (46) (47) (48) (50)) were found iteratively after determining the vehicle masses resulting from the optimization runs and applying sufficient weighting to the constraint errors such that solutions never violated them. The ΔV_{res} is calculated using residual propellant with Eqn. (1). The ΔV_{res} is in units of m/s. The SECO-1 altitude ($H_{\text{SECO,1}}$) is in units of m. The relatively large 15000 m allowable error on $H_{\text{SECO,1}}$ is implemented to

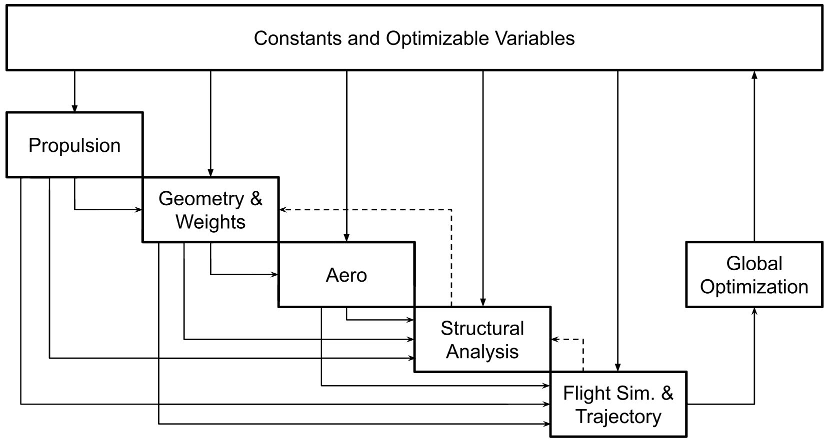

Fig. 5. Global program schematic.

speed up the local-level trajectory convergence. Having smaller tolerance resulted in very long computational run times. The accuracy of the final orbit is unaffected by the inaccurate initial altitude. The initial burn to reach the orbit apogee and the circularization burn can accurately provide the required orbit since they are accomplished by truncating burn times. The staging altitude constraint (Eqn. (49)) ensures the upper stage flight is outside of maximum dynamic pressure ranges, and the load formulation and trajectory formulation assumptions are kept valid. The $\Delta_{\text{H,staging}}$ is in units of m. A 30000 m staging altitude (H_{staging}) constraint is implemented as it was observed to typically be after maximum dynamic pressure. Staging altitudes are later observed, for optimal configurations, to be beyond 50 km above sea level. At 50 km, dynamic pressure is far below the maximum, for the encountered speeds at staging of between two and three km/s. The encountered staging altitudes are driven by dry mass minimization rather than costs from violating staging altitude constraints.

The global optimization studies were conducted for launch vehicles delivering payloads in the range of 150-200 kg to a 500 km polar orbit. The specified orbit is a common standard orbit cited by launch providers. For polar orbits, the vehicle is not affected by the additional velocity due to Earth's rotation and proximity of the launch site to the equator.

To remove an iteration loop and reduce the computational cost, the payload value floats between 150 kg and 200 kg. The nominal payload is set to 150 kg and the residual propellant mass of the second stage is added to provide a net effective payload mass. Assuming the residual propellant can be replaced with payload is conservative as the actual payload mass could take the place of some unneeded structure mass.

The different vehicle configurations investigated in the global optimization studies used the same common feed system for each stage. The feed systems were either entirely backfill-pressure-fed or entirely electric pump-fed feed systems. Initially, pressure-fed configurations were not expected to have sufficient mass margins to support two-stage-to-orbit launch capability, after [17] and [19] presented three-stage vehicles. Initial testing of single-stage sounding rockets, outlined in Ref. [31], showed similar ranges of ΔV between pressure-fed and pump-fed configurations.

The global optimization variables considered are shown in Table 4. The design variables are those that exhibit optima not at extremes, but at mid-ranged values. In addition to the listed global optimization variables, there are three optimization variables for the upper stage trajectory design that are optimized at a local level as previously discussed. There are eight structural section thickness values minimized at a local level for each design iteration, as previously discussed.

Table 4
Global optimization variables.

Symbol	Description
$I_{\text{t,nom},1}$	Total impulse, first stage
$I_{\text{t,nom},2}$	Total impulse, second stage
$t_{b,1}$	Burn time, first stage
$t_{b,2}$	Burn time, second stage
δ_{nozzle}	Nozzle clearance, second stage
$\overline{O/F}_1$	Burn-time-averaged O/F, first stage
$\overline{O/F}_2$	Burn-time-averaged O/F, second stage
$P_{up,1}$	Upstream pressure, first stage
$P_{up,2}$	Upstream pressure, second stage
γ_{po}	Initial pitch over angle for gravity turn
D_{O}	Body outer diameter

The pressurant gas was selected to be helium given its high performance over other inert pressurizing gasses [60]. The helium is modeled to be stored onboard at 300 K and 41.4 MPa. The pressure has little influence on the required mass of pressurant gas, at high initial pressures relative to regulated pressure ($\frac{P_{\text{ox}}}{P_{1,\text{pg}}} << 1$), as seen in Eqn. (18).

The pressurant gas density, assuming ideal gas law, is proportional to pressure. Since the tank mass scaling is proportional to fluid density and inversely proportional to fluid pressure (Eqn. (36)), the pressurant gas vessel mass is approximately constant. The vehicle total inert mass however does decrease for higher pressurant pressures due to the smaller required volume of the rocket upper tank section surrounding structure, which is why a high storage pressure was chosen.

The nozzle exit pressure is desired to be minimized in order to maximize mission-averaged I_{sp} for the investigated mission profile [31] The minimum nozzle exit pressure is 50 kPa in this study. The 50 kPa value meets the Summerfield flow separation criteria (Eqn. (9)) [33] with a 10 kPa margin. The 10 kPa provides margin for exit pressure fluctuations from O/F shift or combustion instabilities. A c^* combustion efficiency of 95% is assumed. High efficiencies have been reported in operational hybrid rocket engine developments [61].

3. Results: global optimization studies

3.1. Particle swarm parameter selection

The individual (c_1), social (c_2), and inertia (w) PSO parameters were first investigated to determine the best-performing values for the launch vehicle design problem. The dual electric pump vehicle concept is selected to investigate the PSO algorithm performance across different PSO parameters. A grid search of the PSO parame-

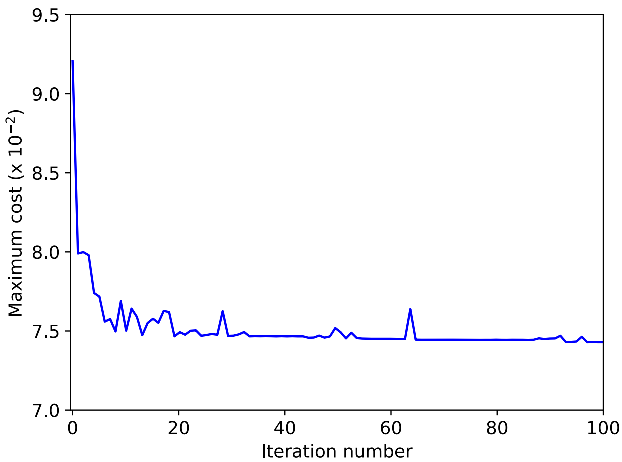

Fig. 6. Electric pump-fed case cost function convergence.

ters was conducted where each parameter took the value of either 0.25, 0.5, or 1.5. Each optimization with a given PSO parameter set was given three attempts. Each attempt used 100 iterations and 10 particles per design dimension. Three attempts were given to observe the effect of the random initial starting conditions and to contrast behavior.

The $[c_1, c_2, w]$ = [1.5, 1.5, 0.5] set resulted in the best consistent cost. All these attempts using [1.5, 1.5, 0.5] converged within 1%. This set of PSO parameters was selected as the best performing and used to generate the optimal design variables for all other investigated designs. The selected PSO parameters also fall within ranges that ensure convergence is met as outlined in Ref. [62].

3.2. Optimal design variables

The best performing PSO parameters ([1.5, 1.5, 0.5]) were used to generate optimal design variables for the investigated vehicle configurations. The optimal design variables and select outputs are shown in Table 5. The cost function value for each iteration for the electric pump-fed case is shown in Fig. 6. Optimal designs for entirely pressure-fed two-stage-to-orbit vehicles were found using relatively low-pressure combustion that had, although not as good, comparable performance to pump-fed systems.

Electric pump-fed systems have higher inert mass payload fractions compared to back-fill systems, however back-fill systems still had comparable performance. The pressure-fed systems used lower combustion pressures. Optimal burn times were in similar ranges for each stage. Upper stage systems had larger burn times due to lower thrust-to-weight requirements. Further investigation into these optimal parameters is conducted in the parametric sensitivity analysis section.

3.3. Parametric sensitivity analysis

The parametric analysis explores the sensitivity of a single design parameter to the delivered inert mass payload fraction (Eqn. (44)). During the parametric studies all other design parameters are held constant. The parametric studies found the design values converged upon by the PSO were the local optimal or within 1%.

3.3.1. Upstream pressure

The upstream pressure, for both stages, is converted to chamber pressure through Eqn. (5). The chamber pressure parametric analysis is presented in Fig. 7.

The pump-fed configuration has a lower sensitivity to changing chamber pressures when compared to the back-fill system, for both stages. The pump-fed system is less sensitive due to the relatively lower increase in mass for increasing chamber pressure. The back-fill system had a higher sensitivity to changing pressures, especially for the upper stage. The back-fill pressure-fed upper stage was found to only be functional in a narrow range of chamber pressures, between approximately 0.1 MPa and 1.0 MPa, outside of this range and the mission was not feasible with two stages.

The first stage had a lower sensitivity to system pressures compared to the upper stage, for both feed system types. The additional mass from higher pressures directly removes payload mass from the second stage, while the effect is less drastic for the first stage.

Since inert mass scales linearly with chamber pressure, as seen in Eqn. (36), while specific impulse is nonlinear with chamber pressure, and has diminishing increase past sufficiently high chamber pressures, the net result is that ΔV is maximized for relatively low system pressures for the considered feed systems. First stage optimal pressure values are higher than second stages due to the required thrust-to-weight ratio. First stage requires higher thrust-to-weight ratios, compared to upper stages, for initial liftoff. Higher thrust-to-weight ratios can be obtained through lower burn times (i.e. higher mass flow) for a given total impulse, or higher chamber pressure.

3.3.2. Burn time

The sensitivity of payload fraction to first stage and second stage burn times is presented in Fig. 8. The inert mass payload fraction is relatively insensitive to first stage burn time ($t_{b,1}$) when compared to other investigated design variables. The payload fraction is particularly insensitive to $t_{b,1}$ past 60 s. The burn time at a fixed total impulse mainly affects the thrust and thrust-to-weight ratio of the launch vehicles. For second stage burn times ($t_{b,2}$), the optimal burn time is desired to be a maximum. Longer upper stage burn times require lower inert mass and better packing efficiency of the hybrid fuel grain as grain length is minimized. The constraint on the upper bound of the upper stage burn time is the failure to reach the desired SECO-1 altitude due to insufficient thrust-to-weight. The lowest possible SECO-1 altitudes are desired

Table 5
Optimal variables of LOx systems with 150-200 kg nominal payload.

Variable [units]	Electric pump	Back-fill
$I_{t,1}$ [MN·s]	19.7	18.2
$I_{t,2}$ [MN·s]	4.5	6.02
$P_{up,1}$ [MPa]	1.69	1.24
$P_{up,2}$ [MPa]	0.780	0.246
$t_{b,1}$ [s]	112	72.5
$t_{b,2}$ [s]	594	596
$\overline{O/F}_1$	2.01	2.07
$\overline{O/F}_2$	2.19	2.21
D_O [m]	1.56	1.51
δ_{nozzle}	0.304	0.177
γ_{po} [°]	84.6	85.3
χ	-0.601	-0.858
c	0.920	0.963
b	0.739	0.570
m_{total} [kg]	12201	12917
$m_{total(stage1)}$ [kg]	10500	10669
$m_{total(stage2)}$ [kg]	1701	2248
$m_{prop(stage1)}$ [kg]	9644	9508
$m_{prop(stage2)}$ [kg]	1351	1846
m_{dry} [kg]	1206	1563
$m_{payload}$ [kg]	162	179
ϵ [%]	13.5	11.4
Length [m]	18.1	21.0

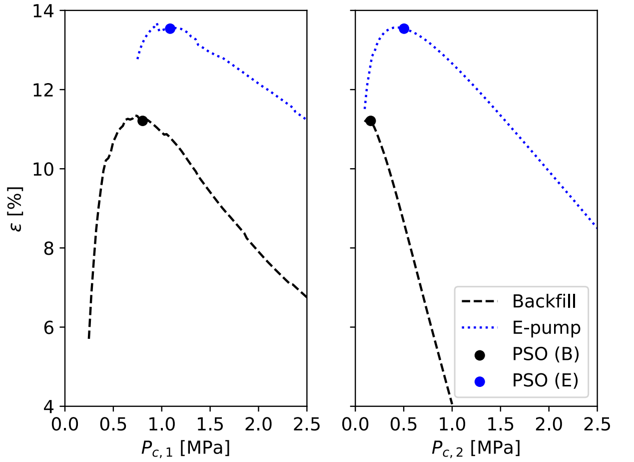

Fig. 7. Payload fraction sensitivity to first stage chamber pressure [left] second stage chamber pressure [right].

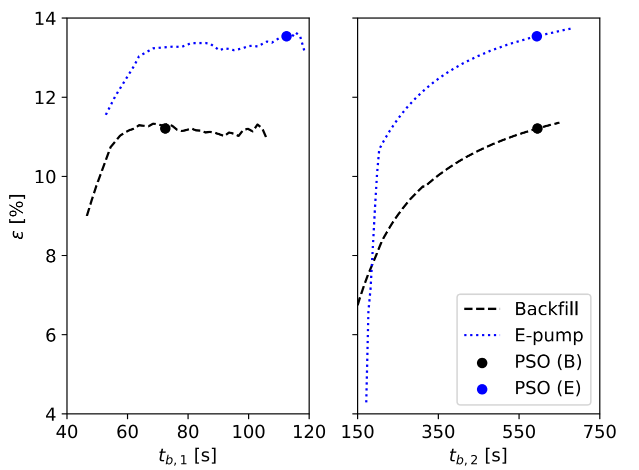

Fig. 8. Payload fraction sensitivity to first stage burn time [left] and second stage burn time [right].

to increase performance. Failure to reach 200 km ±15 km SECO-1 altitude occurs at burn times of 680 s and 650 s for the electric pump and back-fill configurations respectively. Both optimizations converged to approximately 600 s.

3.3.3. Average oxidizer-to-fuel ratio

The results of varying average oxidizer-to-fuel ratio, $\overline{O/F}$, are shown in Fig. 9. For the first stage $\overline{O/F}_1$, the optimal value is 1.9 for both feed systems, which approximately equals the O/F that maximizes I_{sp}. The performance at first stage O/F ($\overline{O/F}_1$) values between 1.9 and 2.2 is approximately constant.

The upper stage back-fill optimal $\overline{O/F}_2$ of 2 matches that predicted by maximizing I_{sp}. The pump-fed case has a slightly higher optimal near 2.2. The pump-fed configuration optimal allocates more propellant mass into the relatively low pressure oxidizer tank to improve the mass efficiency and thus overall performance.

3.3.4. Vehicle diameter and second stage nozzle

The global optimizer always minimized vehicle overall body diameter. It was found that minimal body diameter, within physical geometric constraints, was optimal for the investigated configurations (Fig. 10). The increase in vehicle mass and drag at larger body diameters reduced performance.

The nozzle clearance parameter (δ_{nozzle}), which defines the upper stage nozzle exit diameter via Eqn. (12), has a very smooth relation to performance (Fig. 10). The nozzle clearance parameter

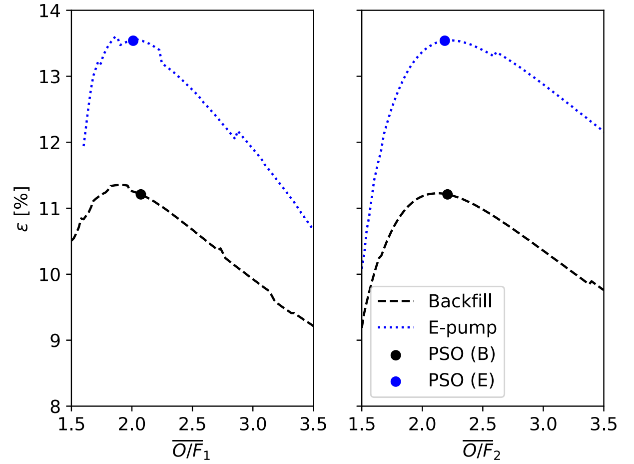

Fig. 9. Payload fraction sensitivity to first stage $\overline{O/F}$ [left] and second stage $\overline{O/F}$ [right].

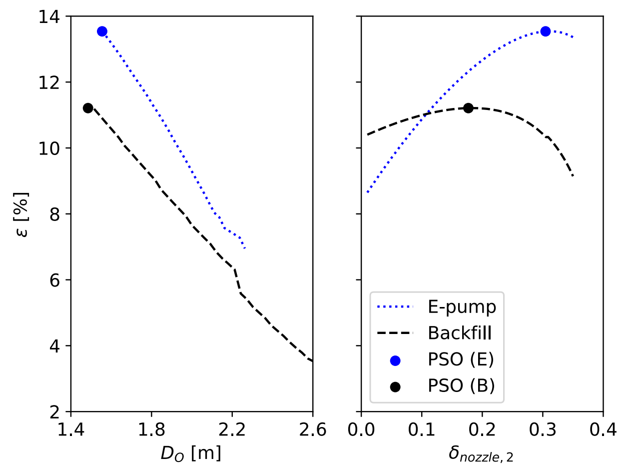

Fig. 10. Payload fraction sensitivity to body diameter [left] and second stage nozzle clearance[right].

trades the I_{sp} by altering nozzle expansion (area ratio), mass of the nozzle bell, and mass of the interstage via the nozzle length.

The back-fill systems optimal δ_{nozzle} is 0.18 while the electric pump-fed system has an optimal value of 0.31. The smaller clearance value (larger nozzle expansion) of the back-fill stages is attributed to the lower chamber pressure which results in lower additional inert mass per increase in performance from higher expansion ratios.

3.3.5. Combustion and structural efficiency

Although combustion efficiency and structural efficiency are not optimization parameters, as they should always be maximized, their sensitivity to the design performance is investigated. The payload fraction sensitivity is investigated for varying $c*$ combustion efficiency (Fig. 11). The payload fraction was found to be highly sensitive to combustion efficiency. The relation was found to linear with similar slope found for the investigated configurations. Payload fractions below 5% are observed below 80% c^* efficiency.

The structural index (I_{str}) is introduced to assess the sensitivity of vehicle performance to the structural efficiency (Fig. 11). The structural index is a factor multiplied to the safety factor of every structural element. It can be viewed as changing each material's strength-to-weight ratio ($\sigma_{allow}/\rho_{material}$) and stiffness-to-weight ratio by the same factor simultaneously. Both vehicle configurations are sensitive to changes in the structural index.

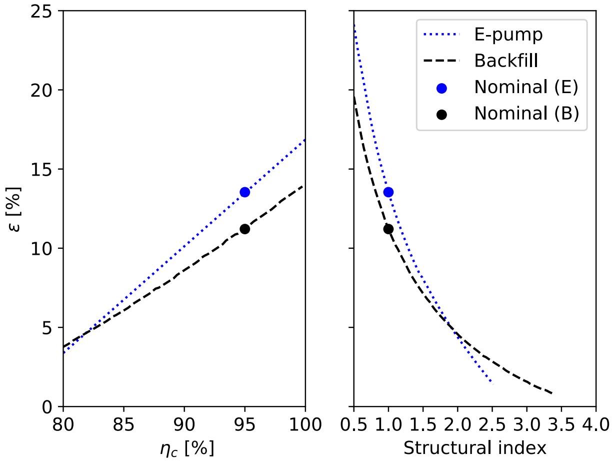

Fig. 11. Payload fraction sensitivity to combustion efficiency [left] and structural efficiency [right].

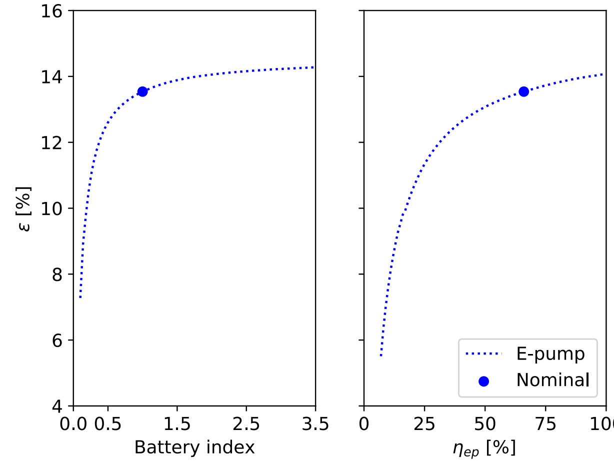

Fig. 12. Payload fraction sensitivity to electric battery index [left] and pump efficiency [right].

Table 6
Local slope at nominal conditions of inert mass payload fraction as a function of system efficiency parameter.

System	$d\epsilon/d\eta_c$	$d\epsilon/dI_{\mathrm{str}}$
Electric pump	0.68	-0.14
Back-fill	0.51	-0.10

The local slopes of the curves in Fig. 11 at nominal conditions are computed and shown in Table 6. The combustion efficiency is the most sensitive design constant. The structural index has low sensitivity considering small changes but has high importance considering it can be more easily improved by choosing materials with a higher strength-to-weight ratio or stiffness-to-weight ratio.

One aspect not considered in this study is the practicality of creating structures with very thin walls (< 1 mm). The pressure-fed designs and the designs with lower structural index provide low inert mass but also present wall thickness values below that conventionally encountered in real designs (< 1 mm). The practicality and cost of manufacturing the vehicle structures and components will be important to include in future studies seeking to minimize vehicle cost.

3.3.6. *Electric pump feed system parameters*

The battery index is introduced here and is a factor multiplied to the battery power density (δ_{bp}) and simultaneously battery energy density (δ_{be}). The batteries are either power-limited or energy-limited during operation, altering both densities at once shows a more meaningful relation of the battery technology. The battery for electric pump-fed vehicle's first stage was found to be power limited while the second stage was found to be energy limited. The battery index's influence on overall performance is shown in Fig. 12. Increasing battery index was found to have diminishing returns past the current level of technology (Battery Index = 1), for the investigated vehicle.

The payload fraction's sensitivity to pump efficiency is presented in Fig. 12. The electric pump efficiency (η_{ep}) was altered between 0.1% and 100%. The payload fraction is initially very sensitive to pump efficiencies. The sensitivity transitions and performance increase become marginal with increasing efficiency past approximately 70%.

4. Conclusions

Two-stage hybrid rocket launch vehicles, delivering 150-200 kg nominal payload to a 500 km polar orbit, were investigated. The vehicles were investigated within a framework considering much of the physics associated with designing orbital-class hybrid rocket vehicles. Two-stage-to-orbit configurations are found to be conceptually viable. Two-stage-to-obit pressure-fed hybrid rocket vehicles had similar performance to electric pump-fed systems. In order to have similar performance to pump-fed systems, the optimal chamber pressure of the pressure-fed systems was lower, most notably for the upper stages.

A PSO parameter grid search was conducted for the present hybrid launch vehicle design problem to tune the PSO parameters. Optimal PSO parameters using high cognitive and social forces and low inertia were found and produced local optimal solutions within 1% which was verified with a parametric study.

Optimal design values and trends were observed in the design studies. Maximizing upper stage burn time maximized inert mass payload fraction. Minimum diameter, within geometric constraints, maximized inert mass payload fraction. Optimal upper stage nozzle expansion ratio is sensitive to the system chamber pressure, which is dependent on the investigated system. Optimal $\overline{O/F}$ is near, but in general not equal to, the value that maximizes I_{sp}. Payload fraction is highly sensitive to combustion efficiency, and to a lesser extent structural efficiency. Payload fraction is approximately linear with combustion efficiency in the ranges investigated. Improving battery energy densities, battery power densities, and pump efficiencies provided marginal system performance increases for the investigated vehicle.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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